



Presented by
BIOFAB Fair >

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Private Event

AI as Designer, Farmer, and CFO for Biomaterials Workshop

SEP

16

Tuesday, September 16

1:00 PM - 2:00 PM



Big Penny Social ↗

London, England

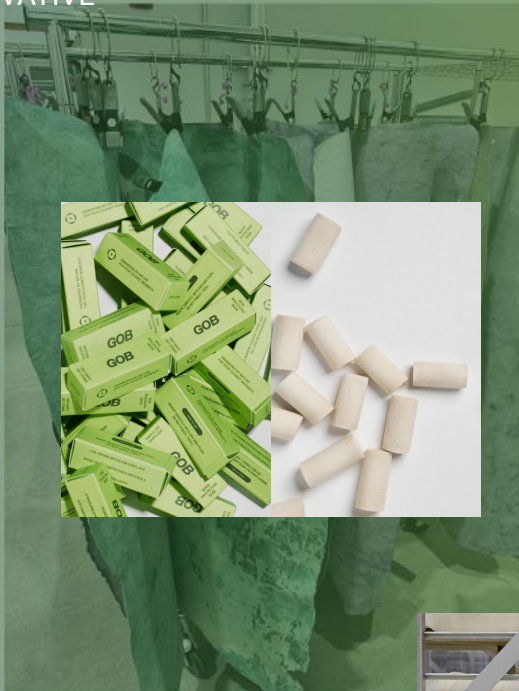
Registration

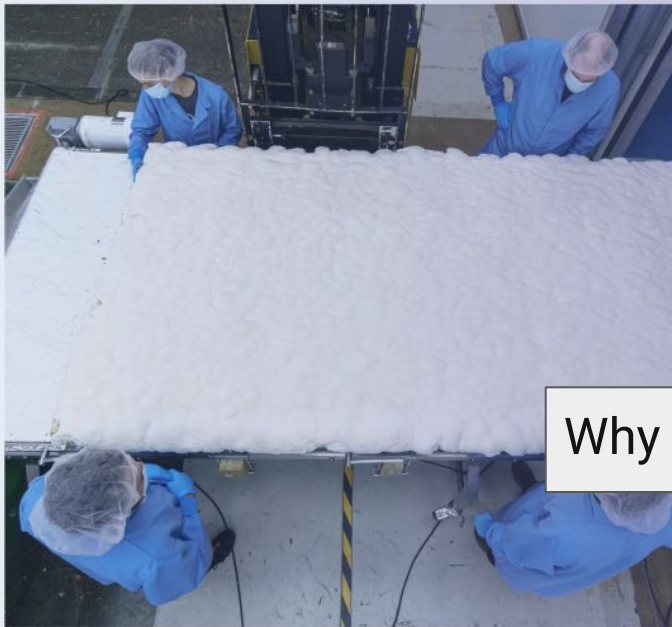


Approval Required

Your registration is subject to approval by the host.

Welcome! To join the event, please register below.





PIVOT-
ABILITY

CAPEX

DESTINY
CONTROL

Why AI matters...

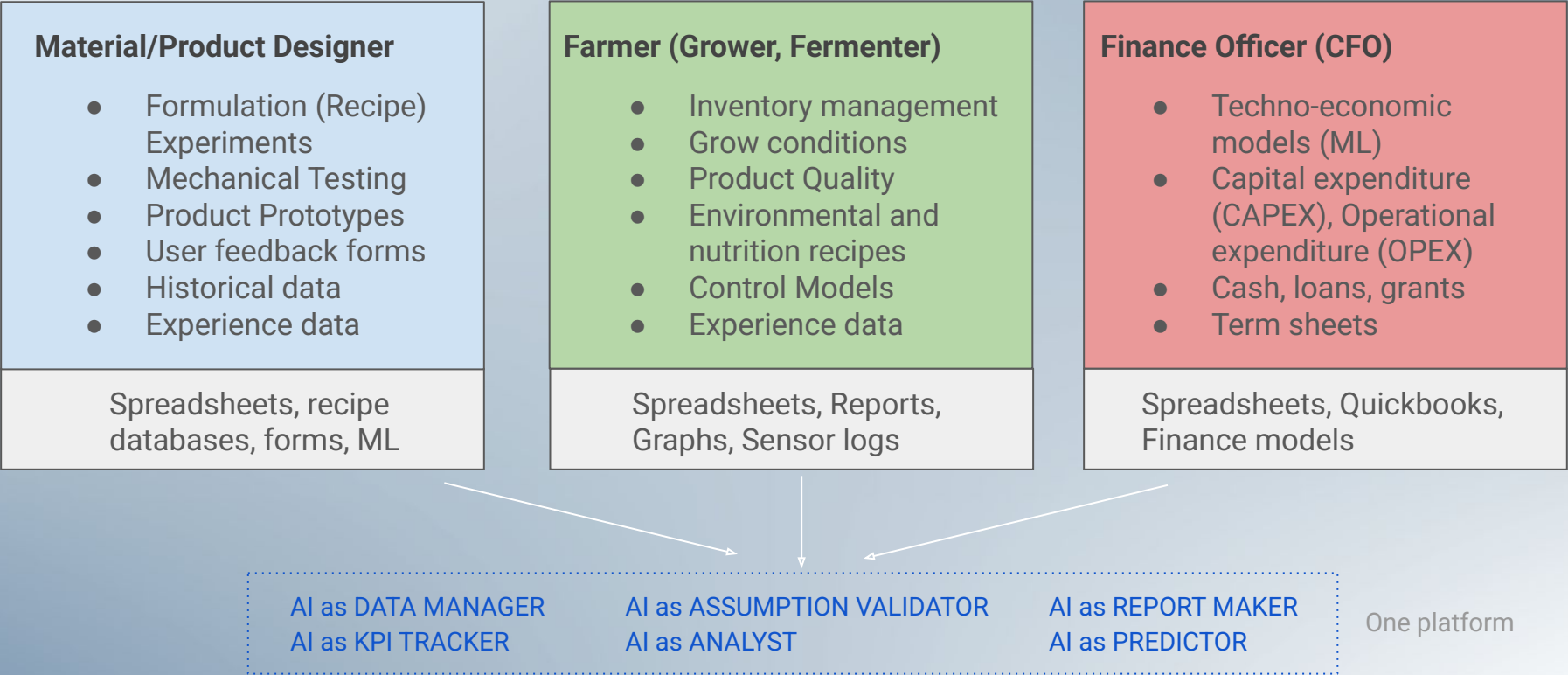
SCALE-READINESS

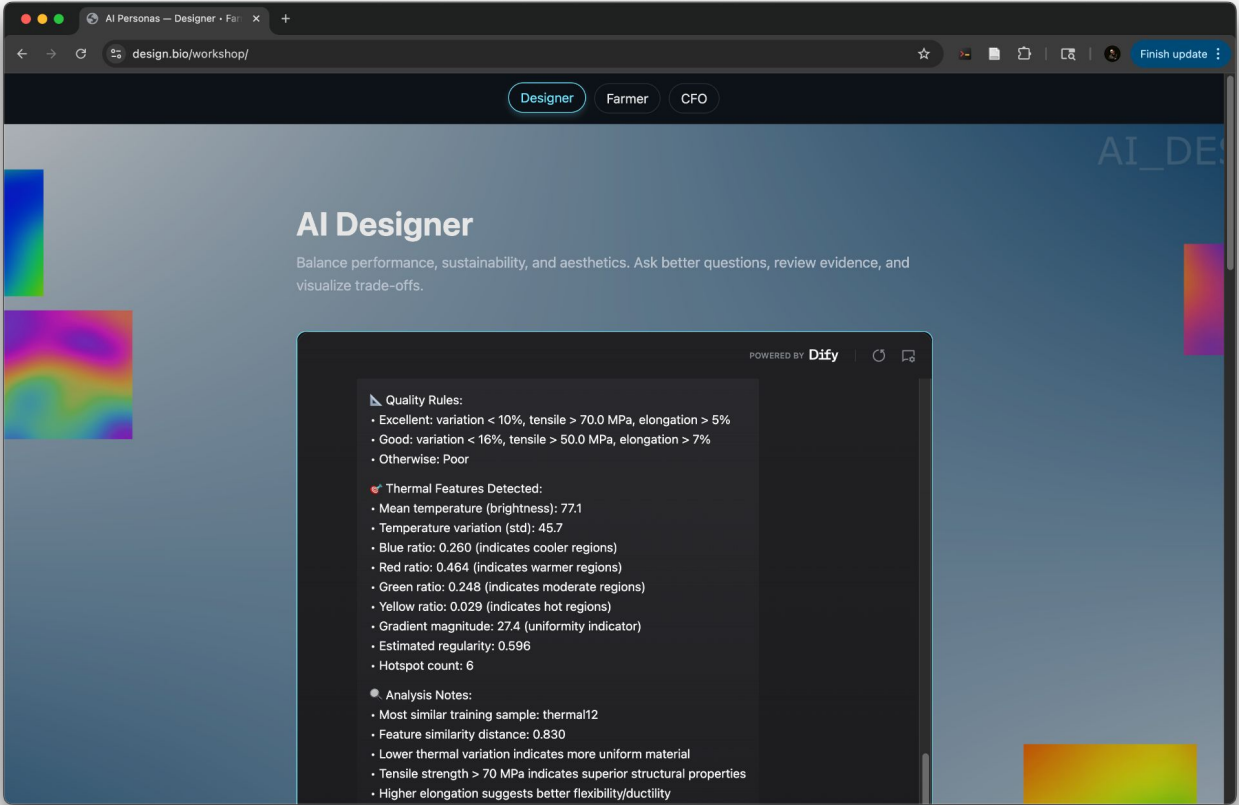


RATE of
LEARNING

AI CFO

DATA vs AI-ASSISTED BUSINESS





Agent driven interaction

AI as Designer

How does AI support material readiness by helping you navigate the trade-offs between performance, sustainability, and aesthetics?

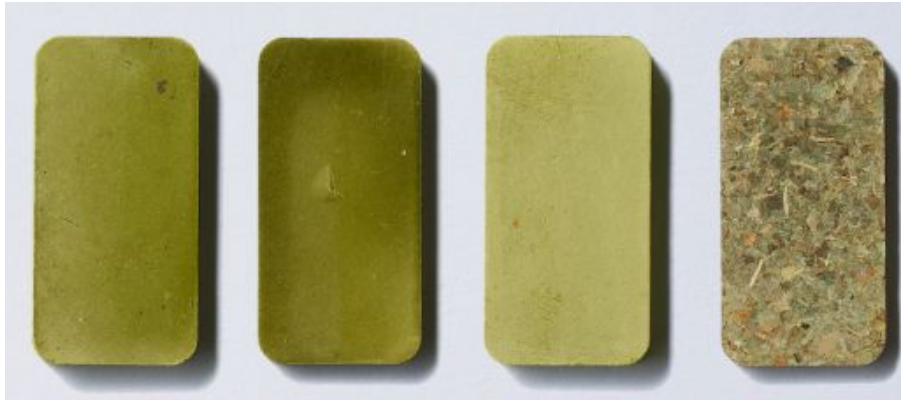
Can you become a KPI-driven designer?





Source: **Fabula** (<https://fabulajp.com/>)

How do you use AI for qualitative experience?

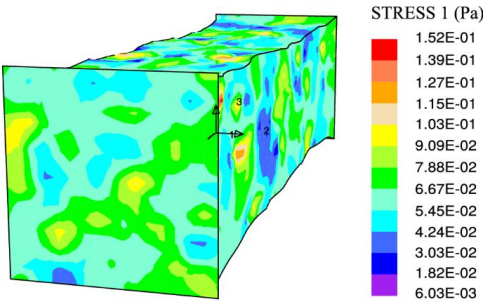


Common questions

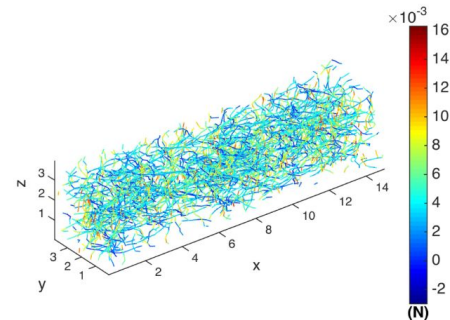
What is the user feedback on our silk finish?

Which finish is liked more silk or matte?

How do you use AI for quantitative assessment?



(a) Stress σ_{11} (Pa) in the matrix



(b) Fiber force f_{fiber} (N) in the network

Source: **Estefaina Cavalie**
<https://class.textile-academy.org/2020>

Jin, T. A computational framework for biomaterials containing three-dimensional random fiber networks based on the affine kinematics. *Biomech Model Mechanobiol* 21, 685–708 (2022).
<https://doi.org/10.1007/s10237-022-01557-6>

How do you quantify design?



Recipe 1

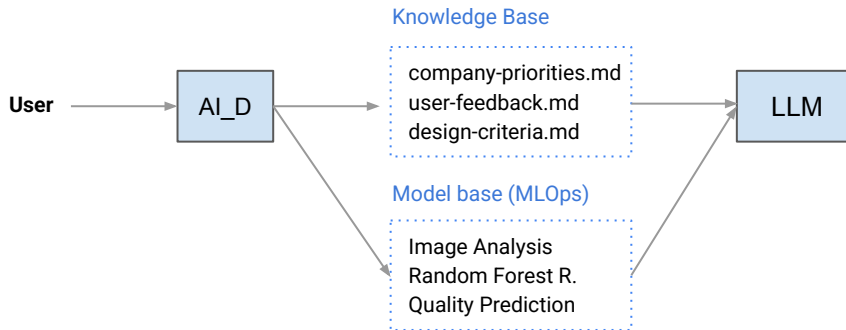


Recipe 2

Common questions

- What is the tensile strength?
- What is elongation at break?
- Abrasion?

AI Designer



Designer

You are in charge of material readiness, but need to know if you meet performance, sustainability criteria.

You are worried about aesthetics. Which interventions will still yield the best aesthetic results

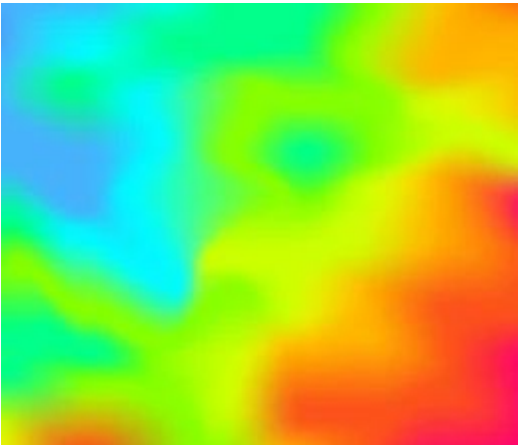
Can you become a KPI-driven designer?

Questions to AI

- What are users saying about the silk finish, especially in terms of durability?
- What is our current user feedback on the matte finish?
- @analyze thermal_t_08.png

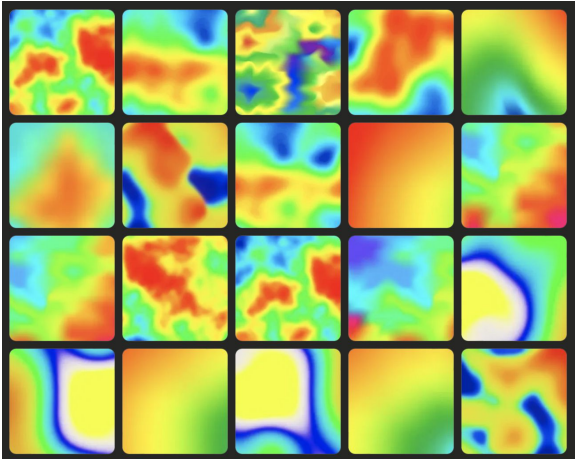
Your Questions?

- ?



thermal_model.pkl

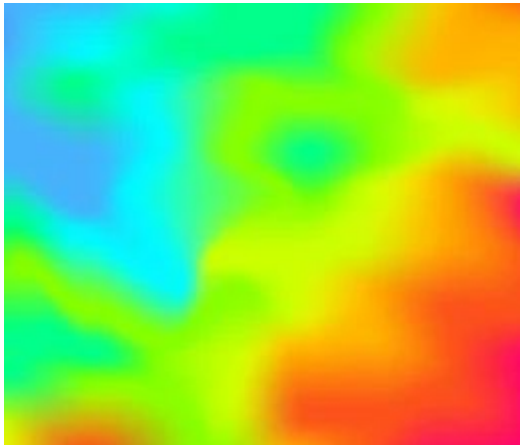
Mean Absolute Error: 14.231
R² Score: 0.913
mean_temp: 0.087
std_temp: 0.075
min_temp: 0.064
max_temp: 0.076
temp_range: 0.127
blue_ratio: 0.127
green_ratio: 0.091
red_ratio: 0.104
yellow_ratio: 0.068
gradient_magnitude: 0.080
uniformity_score: 0.109
hotspot_count: 0.045



Sample ID: **IMG_009**
Regularity Score: **0.85**
Tensile Stregth: **82**
Elongation at Break: **4.1**
Abrasion resistance cycles: **980**
Dominant color zone: **Cool_to_Warm**
Temperature gradient: **Low**
Notes: **Clean linear gradient showing good material consistency**

Demonstrative ML model
trained with 20 test results

thermal_materials_dataset							
Sample_ID	Regularity_Score	Tensile_Strength_MPa	Elongation_at_Break_Percent	Abrasion_Resistance_Cycles	Dominant_Color_Zone	Temperature_Gradient_Intensity	Notes
IMG_001	0.4	45	8.2	850	Mixed_Hot_Cold	High	Complex pattern with sharp hot-cold transitions
IMG_002	0.7	65	6.5	1200	Moderate_Warm	Medium	Good horizontal stratification indicating uniform properties
IMG_003	0.3	35	12.8	750	Hot_Dominant	High	Highly irregular with multiple hot spots
IMG_004	0.8	75	4.2	950	Cool_to_Moderate	Low	Very uniform gradient suggests excellent tensile properties
IMG_005	0.9	85	3.8	1100	Cool_Dominant	Very_Low	Extremely smooth gradient indicates superior strength
IMG_006	0.95	90	3.2	1050	Cool_Dominant	Very_Low	Nearly perfect thermal uniformity
IMG_007	0.2	25	15.6	600	Hot_Mixed	Very_High	Chaotic thermal pattern with poor regularity
IMG_008	0.5	52	7.8	900	Mixed_Zones	Medium	Moderate complexity with distinct thermal zones
IMG_009	0.85	82	4.1	980	Cool_to_Warm	Low	Clean linear gradient showing good material consistency
IMG_010	0.45	48	9.1	820	Warm_Dominant	High	Mixed thermal zones with moderate irregularity
IMG_011	0.45	48	9.1	820	Warm_Dominant	High	Similar pattern to IMG_010 suggesting consistent batch
IMG_012	0.25	28	14.2	650	Hot_Dominant	Very_High	Multiple hot spots indicating material inconsistencies
IMG_013	0.4	45	8.2	850	Mixed_Hot_Cold	High	Similar complexity to IMG_001 thermal signature
IMG_014	0.6	58	7.2	1150	Cool_to_Green	Medium	Good green content suggests strong abrasion resistance
IMG_015	0.75	68	5.8	1300	Cool_with_Artifacts	Medium	Sharp boundaries but overall good uniformity
IMG_016	0.8	72	5.2	1250	Cool_Dominant	Low	Excellent regularity with minimal thermal variation
IMG_017	0.92	88	3.5	1080	Cool_Gradient	Very_Low	Superior linear gradient indicating premium material
IMG_018	0.78	70	5.5	1280	Cool_with_Boundaries	Low	Sharp but organized thermal boundaries
IMG_019	0.88	86	3.7	1090	Cool_Linear	Very_Low	Excellent linear thermal distribution
IMG_020	0.35	38	11.5	780	Hot_Irregular	High	Irregular hot zones suggesting material defects



 Analyzing thermal image: data/experiments/thermal9.png

- =====
- ✓ Image loaded: (221, 258, 3)
 - Mean temperature (brightness): 174.2
 - Temperature variation (std): 24.1
 - Blue ratio: 0.336 (indicates cooler regions)
 - Red ratio: 0.663 (indicates warmer regions)
 - Green ratio: 0.577 (indicates moderate regions)
 - Yellow ratio: 0.000 (indicates hot regions)
 - Gradient magnitude: 8.0 (uniformity indicator)
 - Estimated regularity: 0.040
 - Hotspot count: 7

 MATERIAL PROPERTY PREDICTIONS

- =====
-  Thermal Analysis Results:
- Regularity Score: 0.529/1.000
 - Tensile Strength: 51.7 MPa
 - Elongation at Break: 9.0%
 - Abrasion Resistance: 1,031 cycles
 - Thermal Variation (std): 24.1
 - Thermal Variation (% of 255): 9.4%

 **Quality Assessment: Good**

 Recommendations:
Moderate-stress use; improve uniformity to advance.

-  Quality Rules:
- Excellent: variation < 10%, tensile > 70.0 MPa, elongation > 5%
 - Good: variation < 16%, tensile > 50.0 MPa, elongation > 7%
 - Otherwise: Poor



 Analyzing thermal image: data/test/thermal_t_03.png

- =====
- ✓ Image loaded: (221, 258, 3)
 - Mean temperature (brightness): 133.4
 - Temperature variation (std): 23.7
 - Blue ratio: 0.123 (indicates cooler regions)
 - Red ratio: 0.875 (indicates warmer regions)
 - Green ratio: 0.000 (indicates moderate regions)
 - Yellow ratio: 0.002 (indicates hot regions)
 - Gradient magnitude: 5.1 (uniformity indicator)
 - Estimated regularity: 0.040
 - Hotspot count: 7

 MATERIAL PROPERTY PREDICTIONS

- =====
-  Thermal Analysis Results:
- Regularity Score: 0.450/1.000
 - Tensile Strength: 48.5 MPa
 - Elongation at Break: 10.6%
 - Abrasion Resistance: 930 cycles
 - Thermal Variation (std): 23.7
 - Thermal Variation (% of 255): 9.3%

 Quality Assessment: Poor

 Recommendations:
Reduce variation and increase tensile/elongation.

-  Quality Rules:
- Excellent: variation < 10%, tensile > 70.0 MPa, elongation > 5%
 - Good: variation < 16%, tensile > 50.0 MPa, elongation > 7%
 - Otherwise: Poor



thermal_t_01.png



thermal_t_01.png



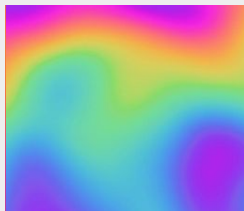
thermal_t_03.png



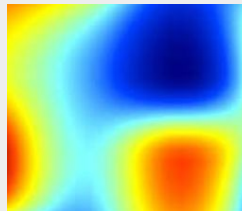
thermal_t_04.png



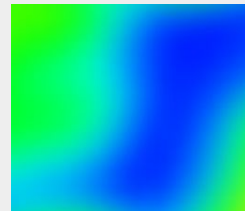
thermal_t_05.png



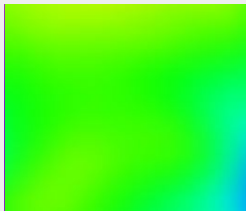
thermal_t_06.png



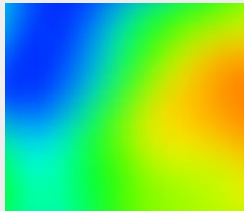
thermal_t_07.png



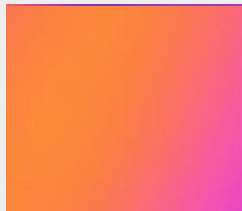
thermal_t_08.png



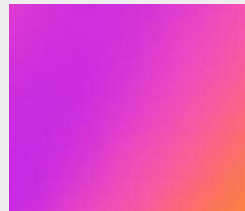
thermal_t_09.png



thermal_t_10.png



thermal_t_11.png



thermal_t_12.png

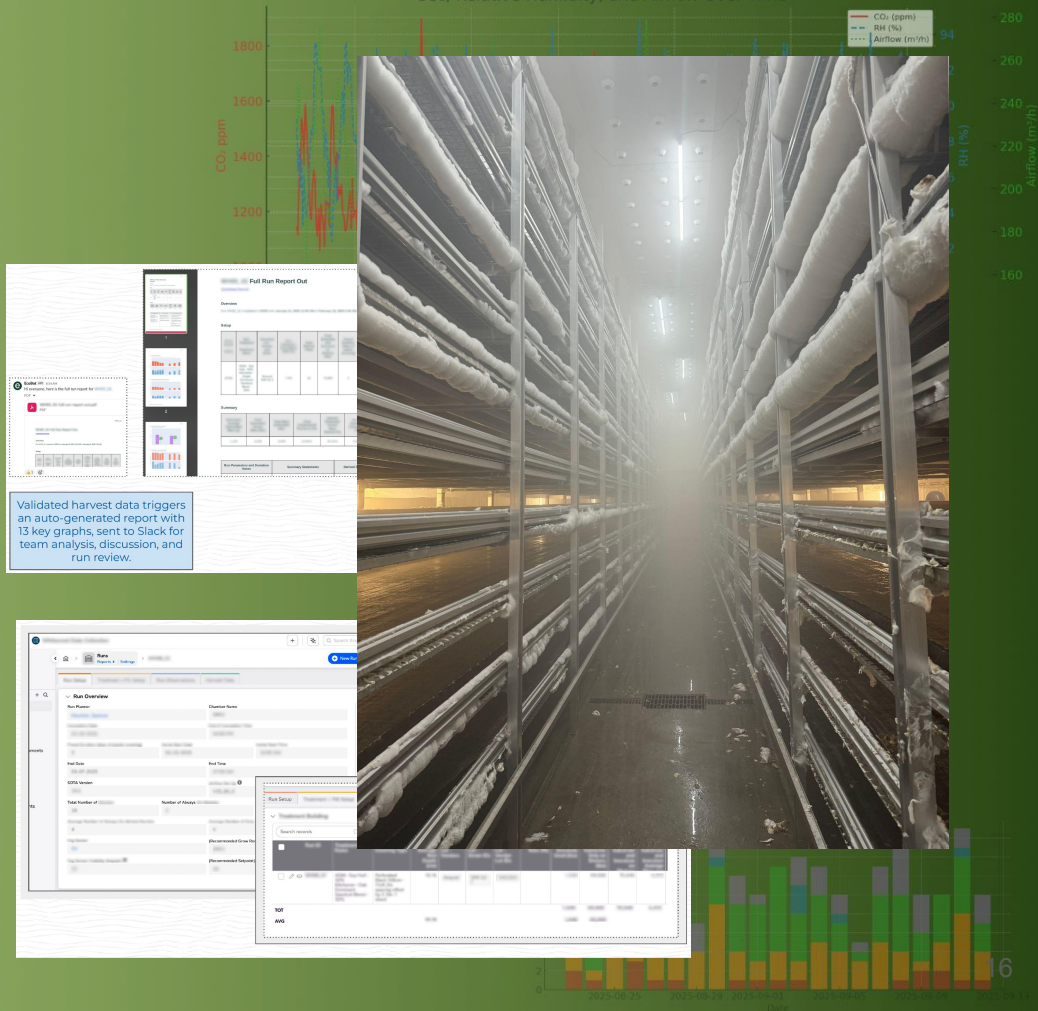
**Test the model
by passing the file
name to the
interface:**

**@analyze
thermal_t_08.png**

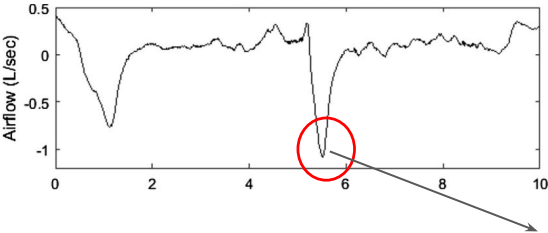
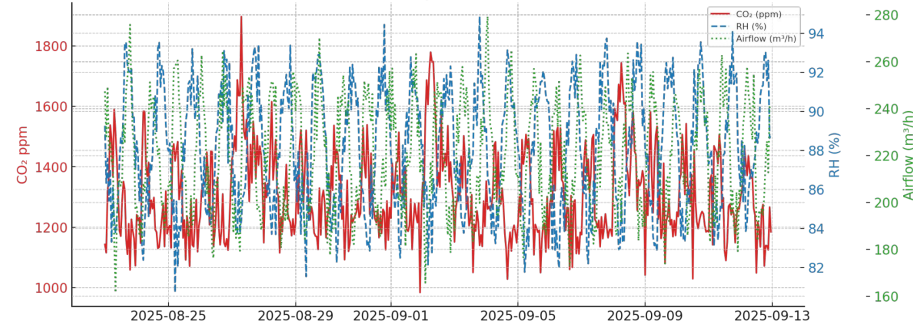
We can build a
version you can pass
images

AI as Farmer (Grower)

- How does AI support scale readiness by predicting quality, yield, and production bottlenecks—and by bringing algorithmic insight together with human intuition?
- What are the key performance indicators of a *material farmer*?



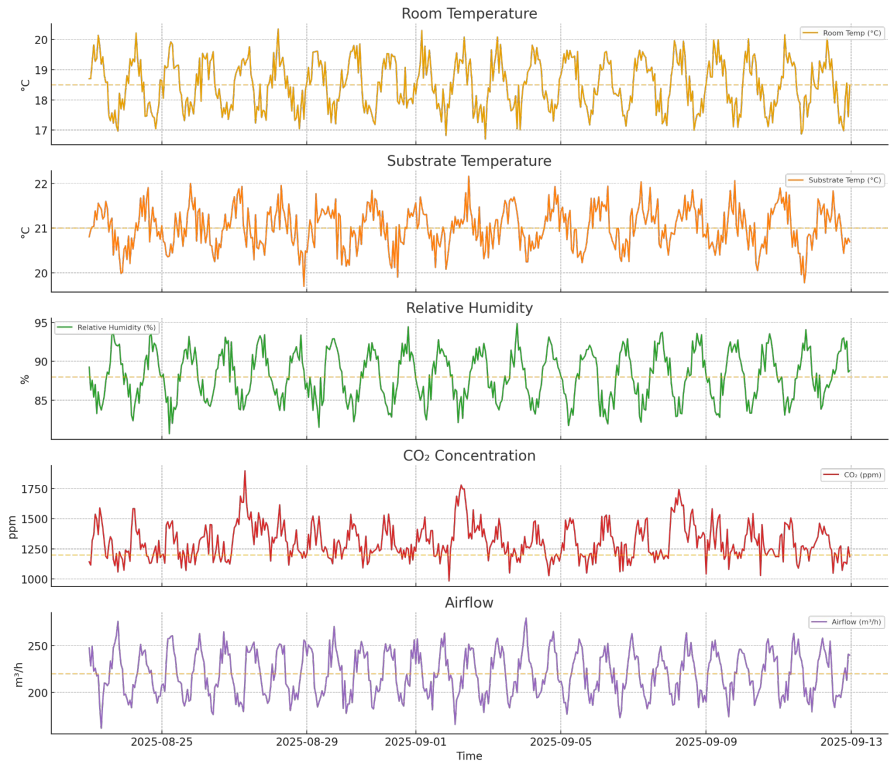
CO₂, Relative Humidity, and Airflow Over Time



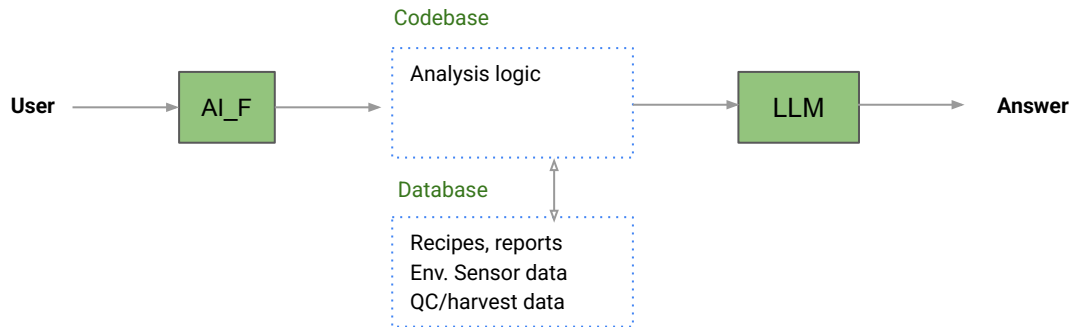
What happened here?

Common issues

- Optimizing recipes
- Finding causes for deviations



AI Farmer



Human Farmer

- There are lots of conditions that affect your growth yield. You would like to know which one affects most
- You try different recipes, you would like to know which one will improve your yield
- You detect an anomaly (i.e., lots of defects in one run), how do you find the root cause?

Questions to AI Farmer

- How much production will we have in 2025
- What was our yield in 2024
- What was our best recipe?
- What was the most influential factor that affect yield?

Your Questions?

- ?

farm_runs_2024																		
run_id	run_period_days	start_date	end_date	recipe	nutrient_input_kg	co2_mbar	humidity_pct	indoor_temp_c	soil_temp_c	grown_area_m2	defects_pct	mass_yield_kg	dev_gas_pct	dev_humidity_pct	dev_indoor_temp_pct	dev_soil_temp_pct	yield_per_m2	avg_deviation_pct
RUN_01	11	2024-01-05	2024-01-16	recipe_2	11.57	1.06	72.5	18.5	18.9	11.19	4.38	16.5	1.62	0.81	0.89	0.56	1.47	0.97
RUN_02	18	2024-01-24	2024-02-11	recipe_1	16.66	1.03	57.3	23.6	20.9	16.9	4.99	24.7	1.33	0.59	1.51	0.84	1.46	1.07
RUN_03	18	2024-02-17	2024-03-06	recipe_3	9.26	1.171	61.7	21.1	19.7	9.18	5.67	13.53	2.05	1.68	0.84	0.56	1.47	1.28
RUN_04	21	2024-03-10	2024-03-31	recipe_1	15.09	1.009	66.7	23.6	21.4	15.87	4.83	23.77	2.11	1.04	1.04	1.15	1.5	1.34
RUN_05	10	2024-04-08	2024-04-18	recipe_3	6.09	0.931	72.7	19.0	20.5	6.43	5.97	8.45	1.65	1.35	1.26	1.22	1.31	1.37
RUN_06	24	2024-04-21	2024-05-15	recipe_2	16.81	0.926	69.0	22.3	22.1	17.4	4.24	27.51	1.12	1.1	1.21	0.78	1.58	1.05
RUN_07	21	2024-05-23	2024-06-13	recipe_3	10.14	0.962	65.1	22.9	21.7	11.07	3.7	15.47	2.1	0.94	0.69	0.87	1.4	1.15
RUN_08	15	2024-06-21	2024-07-06	recipe_2	12.19	0.97	75.4	21.4	22.2	12.38	3.98	18.03	0.66	0.88	0.42	0.76	1.46	0.68
RUN_09	18	2024-07-15	2024-08-02	recipe_1	6.68	1.006	60.8	22.4	21.0	6.58	5.76	10.96	2.94	0.86	0.39	0.97	1.67	1.29
RUN_10	20	2024-08-06	2024-08-26	recipe_2	18.33	0.949	57.6	21.3	18.8	16.96	5.75	26.38	2.07	0.87	0.16	0.49	1.56	0.9
RUN_11	16	2024-08-29	2024-09-14	recipe_2	7.48	0.829	67.1	20.5	20.3	7.96	3.22	12.06	0.95	0.92	1.43	0.96	1.52	1.06
RUN_12	11	2024-09-23	2024-10-04	recipe_3	17.71	1.141	66.0	23.8	20.2	17.62	3.16	25.9	0.43	0.42	0.54	1.75	1.47	0.78
RUN_13	19	2024-10-08	2024-10-27	recipe_1	18.79	0.997	59.8	22.8	20.3	18.47	4.12	27.69	0.4	0.23	0.99	1.53	1.5	0.79
RUN_14	10	2024-11-01	2024-11-11	recipe_1	7.26	1.089	68.4	24.8	21.3	8.15	5.51	11.7	0.4	0.18	1.73	1.59	1.44	0.98
RUN_15	14	2024-11-19	2024-12-03	recipe_2	12.31	0.968	65.0	21.7	21.5	13.09	4.1	21.17	2.77	1.05	1.22	0.08	1.62	1.28
RUN_16	22	2024-12-08	2024-12-30	recipe_3	18.55	0.894	65.2	21.1	20.5	18.95	5.87	27.86	2.4	1.32	1.57	0.84	1.47	1.53
RUN_17	22	2025-01-07	2025-01-29	recipe_1	5.95	1.018	85.0	25.0	19.6	7.53	3.27	9.29	1.24	0.12	0.6	1.03	1.23	0.75
RUN_18	15	2025-02-01	2025-02-16	recipe_2	16.88	0.958	58.0	21.8	18.2	17.43	2.97	28.35	0.18	2.28	0.26	0.46	1.63	0.8
RUN_19	15	2025-02-22	2025-03-09	recipe_2	6.93	0.921	63.0	23.4	21.7	6.78	3.41	10.99	1.16	0.92	0.21	0.8	1.62	0.77
RUN_20	17	2025-03-13	2025-03-30	recipe_1	12.25	1.028	65.4	22.0	19.8	11.83	4.83	17.8	1.78	0.94	1.95	1.53	1.5	1.55

AI as CFO

- How does AI support business readiness by going beyond spreadsheets to model unit economics and market realities?
- Tracking econometrics KPIs
- Baking techno-economic model into chat



You don't really
need to
remember
calculations

```
# --- Calculations ---
util = capacity_utilization_percent / 100.0
mix_f, mix_a, mix_u = fashion_mix_percent/100.0, automotive_mix_percent/100.0,
upholstery_mix_percent/100.0
dr, tax_rate = discount_rate_percent/100.0, tax_rate_percent/100.0

prod_t = production_capacity_tonnes * util
prod_kg = prod_t * 1000.0

avg_price = fashion_price*mix_f + automotive_price*mix_a + upholstery_price*mix_u
var_cost = raw_material_cost + energy_cost + labor_cost + maintenance_cost +
quality_control_cost + packaging_logistics_cost

revenue = prod_kg * avg_price
var = prod_kg * var_cost
gp = revenue - var
gm_pct = (gp/revenue*100.0) if revenue > 0 else 0.0

capex_dollars = initial_capex * 1_000_000.0
fixed_total = (annual_fixed_costs + rd_investment + marketing_sales + corporate_overhead)
* 1_000_000.0
dep = capex_dollars / max(depreciation_period_years, 1)

ebitda = gp - fixed_total
ebit = ebitda - dep
ebt = ebit
taxes = max(0.0, ebt*tax_rate) if ebt > 0 else 0.0
ni = ebt - taxes

rev_kg, fixed_kg = avg_price, (fixed_total+dep)/prod_kg if prod_kg > 0 else 0.0
cost_kg, profit_kg = var_cost + fixed_kg, rev_kg - (var_cost+fixed_kg)

payback = (capex_dollars/ni) if ni > 0 else float("inf")
roi_pct = (ni/capex_dollars*100.0) if capex_dollars > 0 else 0.0

npv = -capex_dollars
for y in range(1, 6):
    npv += ni/((1.0+dr)**y)

rev_f, rev_a, rev_u = revenue*mix_f, revenue*mix_a, revenue*mix_u
```

ASK: I want to analyze a scenario with 80% capacity utilization, automotive price at \$32/kg, and raw material costs at \$9/kg. Use defaults for everything else."

 Fiber Manufacturing Financial Analysis

I'll help you analyze the financial performance of your fiber manufacturing operation. Please provide the following information (you can use the default values as a starting point):

 PRODUCTION PARAMETERS

- **Production Capacity:** What's your annual production capacity? (default: 60 tonnes)
- **Capacity Utilization:** What percentage of capacity are you currently using? (default: 75%)

 PRICING BY SEGMENT (per kg)

- **Fashion Segment:** Current selling price (default: \$20.00/kg)
- **Automotive Segment:** Current selling price (default: \$30.00/kg)
- **Upholstery Segment:** Current selling price (default: \$25.00/kg)

 PRODUCT MIX (percentages should total 100%)

- **Fashion Mix:** What % of production goes to fashion? (default: 40%)
- **Automotive Mix:** What % goes to automotive? (default: 35%)
- **Upholstery Mix:** What % goes to upholstery? (default: 25%)

 VARIABLE COSTS (per kg)

- **Raw Materials:** Cost per kg (default: \$8.00)
- **Energy:** Cost per kg (default: \$3.50)
- **Labor:** Cost per kg (default: \$2.50)
- **Maintenance:** Cost per kg (default: \$1.50)
- **Quality Control:** Cost per kg (default: \$1.00)
- **Packaging & Logistics:** Cost per kg (default: \$2.00)

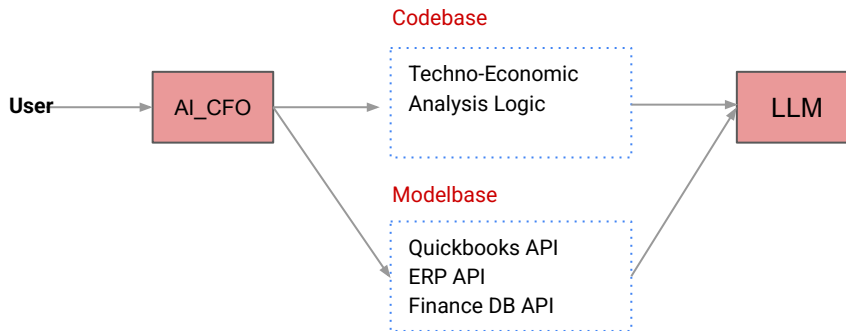
 FIXED COSTS & INVESTMENT (in millions)

- **Initial CAPEX:** Total capital investment (default: \$2.5M)
- **Annual Fixed Costs:** Facilities, insurance, etc. (default: \$0.4M)
- **R&D Investment:** Annual R&D spending (default: \$0.15M)
- **Marketing & Sales:** Annual budget (default: \$0.1M)
- **Corporate Overhead:** Annual overhead (default: \$0.2M)

 FINANCIAL ASSUMPTIONS

- **Depreciation Period:** Equipment life in years (default: 10 years)
- **Discount Rate:** Cost of capital (default: 12%)
- **Tax Rate:** Corporate tax rate (default: 25%)

AI CFO



Human CFO

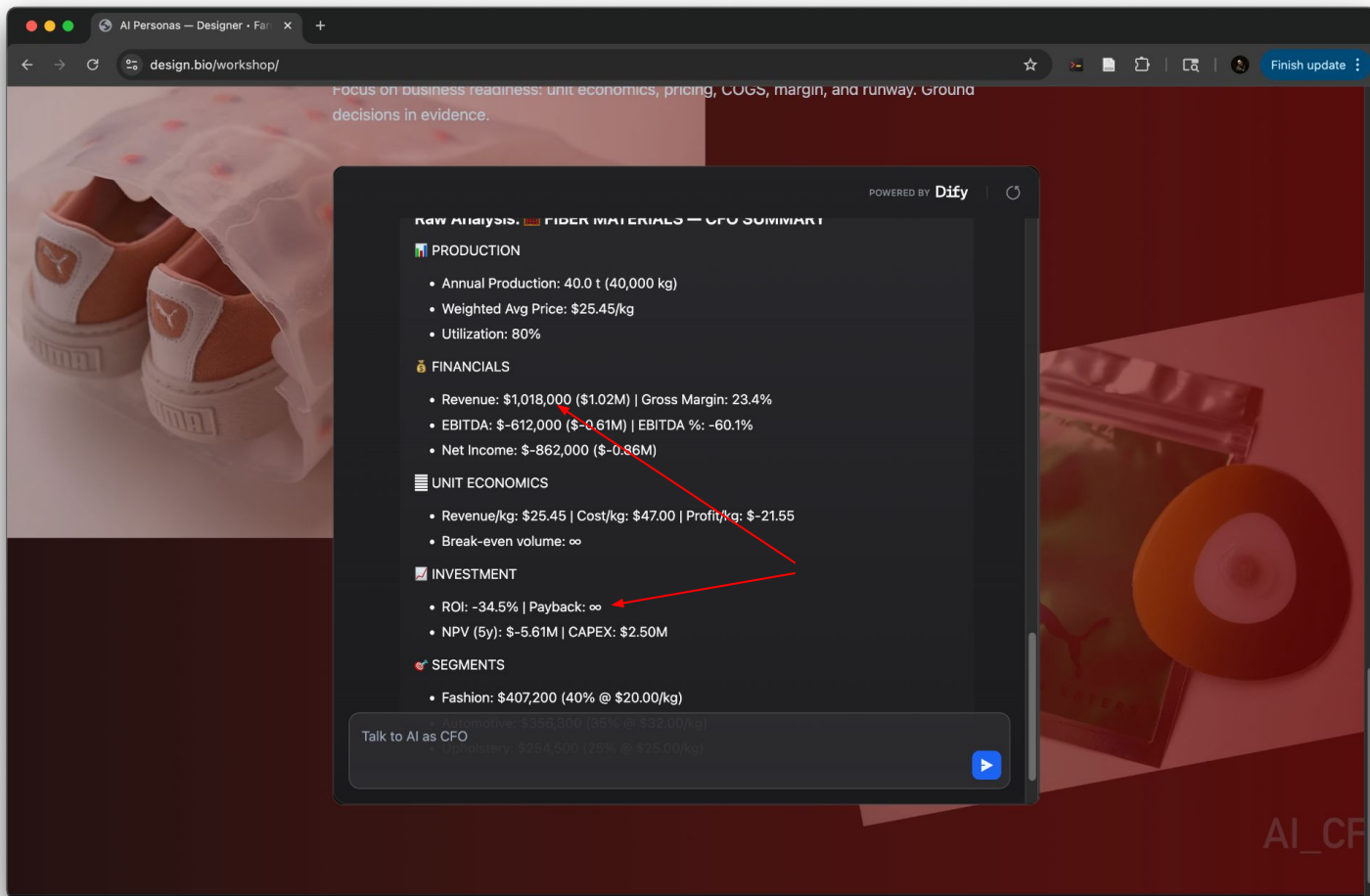
- What is our burn-rate? Where is the money spent?
- Where are our bottlenecks (i.e., supply-side or demand-side)
- What is ROI expectations?
- What are our capacity requirements?

Questions to AI CFO

- I want to analyze a scenario with 80% capacity utilization, automotive price at \$32/kg, and raw material costs at \$9/kg. Use defaults for everything else.
- Capacity 80%, raw \$9, energy \$3.6, fashion price \$21, automotive price \$32, mix 45/35/20. What's EBITDA and profit/kg?

Your Questions?

- ?



Thank you

orkan@design.bio | **Orkan Telhan, PhD**